

Quick Reference (what students saw)

Multiplicity odd $\Rightarrow$ cross; even $\Rightarrow$ bounce. Even: $f(-x) = f(x)$; Odd: $f(-x) = -f(x)$.
End behavior from leading term ax^n . Max extrema $\leq n - 1$, max inflections $\leq n - 2$.
 $(a + b)^n = \sum \binom{n}{k} a^{n-k} b^k$.

1. (6 points) **Solution:** Roots: 4 (mult 2, bounce), -1 (mult 3, cross), 2 (mult 1, cross).

2. (6 points) **Solution:** Monic: $(x + 2)^2 \cdot x \cdot (x - 3)^3$.

3. (6 points) **Solution:** Bounce at $-1 \Rightarrow$ even mult (min 2). Cross at 2 $\Rightarrow$ odd mult (min 1). No other real roots. Smallest degree $= 2 + 1 = 3$.

4. (6 points) **Solution:** (a) $f(-x) = -x^5 + x^3 = -(x^5 - x^3) \Rightarrow$ odd.
(b) $g(-x) = x^4 - 2x^2 + 1 = g(x) \Rightarrow$ even.
(c) $h(-x) = -x^3 + x^2 \neq \pm h(x) \Rightarrow$ neither.

5. (6 points) **Solution:** Even degree 4 with roots ± 3 : factor $(x^2 - 9)$. Let $f(x) = A(x^2 - 9)^2 = A(x^4 - 18x^2 + 81)$. y -int $= f(0) = A \cdot 81 = 18 \Rightarrow A = \frac{2}{9}$. Thus $f(x) = \frac{2}{9}(x^2 - 9)^2$.

6. (6 points) **Solution:** (a) $-5x^7$: odd, $a < 0 \Rightarrow$ left up, right down.
(b) $2x^6$: even, $a > 0 \Rightarrow$ both up.
(c) $-x^4$: even, $a < 0 \Rightarrow$ both down.

7. (6 points) **Solution:** Ends opposite $\Rightarrow$ odd degree. Left down/right up $\Rightarrow a > 0$ (positive leading coefficient).

8. (6 points) **Solution:** Touches and turns (bounce) $\Rightarrow$ multiplicity even.

9. (6 points) **Solution:** x -ints: $x = 2$ (mult 2, bounce), $x = -4$ (mult 1, cross). y -int: $f(0) = (0 - 2)^2(0 + 4) = 4 \cdot 4 = 16$.

10. (6 points) **Solution:** Max extrema $= 7 - 1 = 6$. Max inflections $= 7 - 2 = 5$.

11. (8 points) **Solution:** (a) Degree $= 2 + 3 = 5$, leading term $\sim x^5$.
(b) Odd, $a > 0 \Rightarrow$ left down, right up.
(c) $x = -2$ (mult 2, bounce), $x = 1$ (mult 3, cross with flatten).
(d) $P(0) = (0 + 2)^2(0 - 1)^3 = 4 \cdot (-1) = -4$.

12. (8 points) **Solution:** $Q(x) = -2(x - 4)^2(x + 1)$. Degree $= 3$, leading term $-2x^3$ (odd, $a < 0$).
End behavior: left up, right down.
 x -ints: $x = 4$ (mult 2, bounce), $x = -1$ (mult 1, cross).
 y -int: $Q(0) = -2(16)(1) = -32$.

13. (6 points) **Solution:** Multiplicity 3 at $x = 2$ implies $p(2) = 0$, $p'(2) = 0$, $p''(2) = 0$ (but generally $p^{(3)}(2) \neq 0$).

14. (6 points) **Solution:** Multiplicity 3 (odd) crosses and often flattens with an S-shape; 1 crosses with no flatten; 2 would bounce. So $m = 3$.

15. (6 points) **Solution:** Degree $= 4 + 1 = 5$. Leading coeff: $1 \cdot 2 = 2 > 0$ (from highest powers: $(x^4)(2x)$). End behavior (odd, $a > 0$): left down, right up.

16. (6 points) **Solution:** Current multiplicities: $2 + 1 + 2 = 5$. Degree 6 total $\Rightarrow$ remaining multiplicity = 1 (a single simple root).

17. (6 points) **Solution:** $(x+2)^3 = \binom{3}{0}x^3 + \binom{3}{1}x^2(2) + \binom{3}{2}x(2^2) + \binom{3}{3}2^3 = x^3 + 6x^2 + 12x + 8$.

18. (6 points) **Solution:** $(2x - 3)^6 = \sum_{k=0}^6 \binom{6}{k}(2x)^{6-k}(-3)^k$. We want $x^4 \Rightarrow 6 - k = 4 \Rightarrow k = 2$. Coeff = $\binom{6}{2}2^4(-3)^2 = 15 \cdot 16 \cdot 9 = 2160$.

19. (6 points) **Solution:** $(x + 1)^8$ has $n = 8$ terms; middle is $k = 4$: $\binom{8}{4}x^4 1^4 = 70x^4$.

20. (10 points) **Solution:** $S(x) = -x(x + 1)^2(x - 3)^2$. Degree = $1 + 2 + 2 = 5$ (odd).
Leading coefficient negative (top term $\sim -x^5$).
End behavior (odd, $a < 0$): left up, right down.
 x -ints: $x = 0$ (mult 1, cross), $x = -1$ (mult 2, bounce), $x = 3$ (mult 2, bounce).
 y -int: $S(0) = 0$ (at the origin).